PROJECT REPORT

INTERMEDIARY STATISTICAL MODELING FOR ANALYTICS

TOPIC: HOUSING PRICE PREDICTION

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INTRODUCTION

The study of home price forecasting is at the forefront of real estate economics and is driven by the fundamental requirement to understand and predict changes in property prices. Over time, in an attempt to solve this persistent issue, scholars and practitioners have been pushed to take up new initiatives due to a major research vacuum in this sector.Despite substantial attempts to develop trustworthy forecasting models, the field is still having difficulty accurately anticipating future property values. Although past efforts have yielded differing degrees of success, they have been unable to attain the degree of precision and consistency that real estate market participants desire.

This research gap in housing price forecasting is not just a theoretical problem; it also has practical ramifications for legislators, investors, homeowners, and other stakeholders. Accurate housing market developments estimates are necessary to make well-informed decisions about financial planning, urban development, real estate investment, and economic policy formation.

Closing this disparity requires utilizing the King County, Washington, dataset, which is renowned for its wealth of data offering insights into the factors influencing local property prices. However, it's likely that past attempts abused this data, opening the door for more exhaustive investigation and optimization.

This research tries to overcome past limitations and enhance forecast performance by incorporating lessons learned from earlier efforts. Important methods include ensemble model fine-tuning, advanced feature engineering, and outlier resolution. The principal aim is to develop a model that not only accurately forecasts property prices but also demonstrates robustness and dependability across various market conditions.

As a hub for real estate activity and diversity, King County presents a unique opportunity to enhance home price prediction models. Because the market is so complex, it is essential to develop models that adapt to changing conditions. Because of this, this study focuses on King County to create a model that is tailored to meet the demands of this specific market and offers a model for enhancing forecasting skills in other intricate real estate markets.

In conclusion, this new project is motivated by the need to close the existing research gaps. The objective is to create a more accurate home price prediction model that will beat existing methods and set the bar for more research in the field by leveraging the large dataset from King County, Washington, and the experience from past attempts.

RESEARCH QUESTION

Is home price estimation still untapped despite successful prior results and extensive dataset utilization, regression techniques, and model adjustments?

Reasons for selecting this problem question:

This specific problem question was selected because it addresses a significant gap in the literature about the projection of home prices. Despite previous attempts, existing models still cannot predict housing prices with any degree of reliability, indicating that further research is required to identify the reasons behind these problems.

What the problem question will address is:

The objective of the problem question is to pinpoint the factors that contribute to the inadequacies of the models that are currently in use for housing price calculation. It examines potential shortcomings in previous approaches, including feature engineering techniques, model tuning, and outlier management, and assesses how these factors affect the predictive power of the models.

Why this is important:

Improving the accuracy and reliability of models used to predict property prices requires resolving this issue. By understanding the factors affecting the performance of existing models, researchers can develop more effective ways to overcome these limitations and enhance anticipated performance. This has significant implications for people who rely on accurate housing price estimates to inform their actions, including legislators, urban planners, homeowners, and real estate investors.

Is it worth investigating?

There is much to be gained by looking into this issue, including potential ways to improve the effectiveness and efficiency of home price prediction models. By resolving identified restrictions and enhancing prediction performance, researchers can make better decisions in the real estate market that will benefit both individuals and communities. Researching this matter may also result in advancements in the field of housing market forecasting, which will benefit academics and practitioners alike.

Importance in today's market:

1. Well-Informed Decision Making: People and organizations can make educated decisions regarding the purchase, financing, sale, or investment of real estate by using accurate house price estimates. Stakeholders can optimize real estate investments, avoid risks, and maximize rewards by comprehending future pricing patterns.

2. Economic Stability: The state of the economy as a whole is greatly influenced by the real estate market. Changes in house prices may have an effect on borrowing practices, spending patterns, investment choices, and consumer confidence. Therefore, by lowering market volatility and uncertainty, raising the accuracy of home price forecast models can help to promote improved economic stability.

3. Risk management: Housing price predictions are necessary for investors and financial institutions to manage risk in an efficient manner. Precise forecasts aid in identifying possible hazards related to mortgage funding, real estate investment portfolios, and the securitization of mortgage-backed securities. Stakeholders may lower risks and prevent losses by implementing measures that are based on trustworthy estimations.

4. Innovation and Research: Building a link between the domain of home price forecasting and other domains of knowledge leads to improvements in predictive analytics, machine learning, and data science. Researchers enhance predictive modeling systems by exploring new approaches and pushing the limits of expected accuracy. This helps not just the real estate industry but also other sectors that depend on data-driven decision-making.

Overall, in order to support urban development, promote economic stability, facilitate informed decision-making, lower risks, increase market competitiveness, and foster innovation in predictive analytics, it is imperative to improve housing price prediction models based on the problem and motivation outlined above.

DATA ANALYSIS

Importing Python Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn import linear_model,metrics
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
```

Loading Dataset

```
data = pd.read_csv('kc_house_data.csv')
```

```
data.head()    #return first 5 tuples of the data set
```

	id	date	price	bedrooms	bathrooms	sqft_living	sqft_lot	floors	waterfront	view	...	grade	sqft_above	sqft_basement	yr_built	yr_renovated	zipcode
0	7129300520	20141013T000000	221900.0	3	1.00	1180	5650	1.0	0	0	...	7	1180	0	1955	0	98178
1	6414100192	20141209T000000	538000.0	3	2.25	2570	7242	2.0	0	0	...	7	2170	400	1951	1991	98125
2	5631500400	20150225T000000	180000.0	2	1.00	770	10000	1.0	0	0	...	6	770	0	1933	0	98028
3	2487200875	20141209T000000	604000.0	4	3.00	1960	5000	1.0	0	0	...	7	1050	910	1965	0	98136
4	1954400510	20150218T000000	510000.0	3	2.00	1680	8080	1.0	0	0	...	8	1680	0	1987	0	98074

5 rows × 21 columns

```
data.describe()  # to perform some statistical operations on data
```

```
data.describe()
```

	age	Customer_Reviews_Importance	Personalized_Recommendation_Frequency	Rating_Accuracy	Shopping_Satisfaction
count	602.000000	602.000000	602.000000	602.000000	602.000000
mean	30.790698	2.480066	2.699336	2.672757	2.463455
std	10.193276	1.185226	1.042028	0.899744	1.012152
min	3.000000	1.000000	1.000000	1.000000	1.000000
25%	23.000000	1.000000	2.000000	2.000000	2.000000
50%	26.000000	3.000000	3.000000	3.000000	2.000000
75%	36.000000	3.000000	3.000000	3.000000	3.000000
max	67.000000	5.000000	5.000000	5.000000	5.000000

Interpretation of the data

- It appears that each attribute has a distinct identity.
- The average price of a property is $540,088; the standard variation is $367,127. The pricing range for it is $75,000 to $7,700,000.
- The average number of bedrooms is 3.37; the lowest is 0, and the highest is 33.
- The average number of bathrooms is 2.11; the lowest and greatest values, accordingly, are 0 and 8. With an average of 2,079.9 square feet, the living area varies from 290 to 13,540 square feet.
- The average lot size is 15,107 square feet, with a range of 520 to 1,651,359 square feet.
- 1.49 is the average, 3.5 is the highest, and 1 is the lowest.
- This variable, which indicates whether or not the property has a waterfront (1) or (0), appears to be binary.
- This could indicate the number of times the property has been inspected. Greater square footage: It has a range of 0–4. This could be the total grade of the dwelling unit, which ranges from 1 to 13.
- Condition, this could be a helpful tool to gauge the general condition of the house on a scale of 1 to 5. 1,788.4 square feet is the average amount of living area above ground.
- The average size of a basement is 291.5 square feet.
- The oldest house was built in 1900, and the newest was built in 2015. Usually, the construction year is 1971.
- Year Renovated, given that it takes an average of 84.4 years to renovate a property, the majority of properties have not undergone renovations. The homes are dispersed throughout multiple zip codes, which span from 98001 to 98199.
- The residence's latitude and longitude coordinates.
- The area of the residential building on the plot of land is fifteen square feet. 15. The average lot and living space measurements of the fifteen nearest properties may be as follows.

Data Cleaning

```
data.shape #returns dimensions of data set i.e no.of rows and columns
```

(21613, 21)

```
data.columns # returns names of all columns
```

```
Index(['id', 'date', 'price', 'bedrooms', 'bathrooms', 'sqft_living',
       'sqft_lot', 'floors', 'waterfront', 'view', 'condition', 'grade',
       'sqft_above', 'sqft_basement', 'yr_built', 'yr_renovated', 'zipcode',
       'lat', 'long', 'sqft_living15', 'sqft_lot15'],
      dtype='object')
```

```
data_redefined = data[['price','bedrooms', 'bathrooms', 'sqft_living', 'sqft_lot',
                       'floors','waterfront','view', 'condition',
                       'grade', 'sqft_above','sqft_basement',
                       'yr_built','yr_renovated','zipcode', 'lat',
                       'long','sqft_living15', 'sqft_lot15']]
# New data frame containing Explanatory variables and Response variable price after removing id and date
```

As you can see we have removed the id and date column from our dataset because it wasn't helping us in our analysis in any way

Then we checked for any null values in our dataset but there weren't any null values in our dataset.

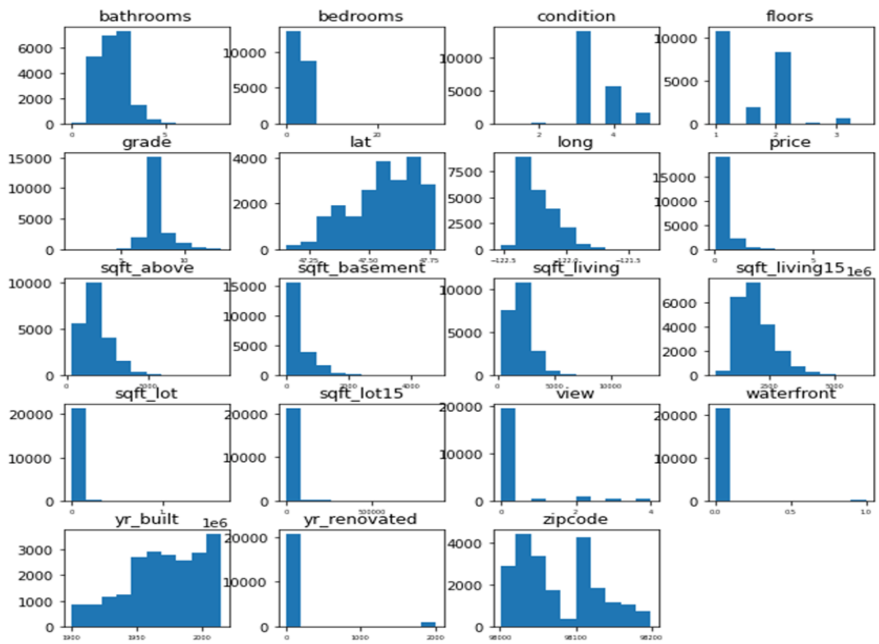
```
data_redefined.isnull().sum()  # checking for any null values
```

```
price            0
bedrooms         0
bathrooms        0
sqft_living      0
sqft_lot         0
floors           0
waterfront       0
view             0
condition        0
grade            0
sqft_above       0
sqft_basement    0
yr_built         0
yr_renovated     0
zipcode          0
lat              0
long             0
sqft_living15    0
sqft_lot15       0
dtype: int64
```

DATA VISUALIZATION

Frequency distribution of the attributes of dataset:

```
data_redefined.hist(figsize=(10,10),grid=False,xlabelsize=5)
```



- Each histogram shows a different property attribute.
- The x-axis varies according to the attribute (number of bathrooms, latitude, etc.).
- On the y-axis is the frequency or count of qualities with specific values.

As an illustration:

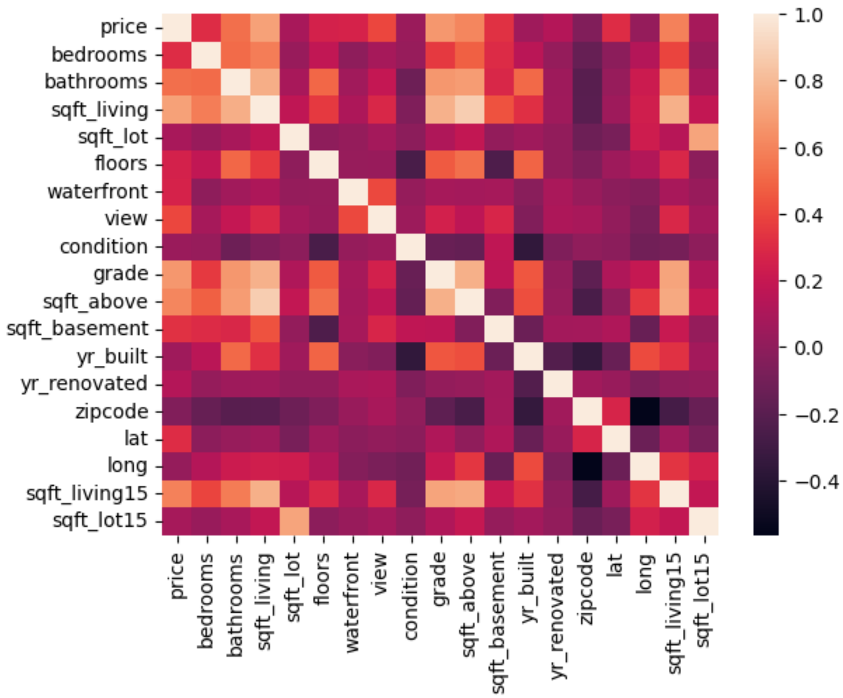
- The "Bathrooms" histogram shows that most residences have one or two bathrooms.
- A grouping of properties at a lower price range is shown by a spike at one end of the "Price" histogram.
- Geographic data (latitude and longitude) histograms display a more homogeneous distribution.
- The "Year Built" and "Year Renovated" histograms show an increase in the last few years.

Heatmap of coefficients correlation

```
# plotting the histogram for entire dataset and see frequency distribution of variables

correlation=data_redefined.corr()  # finding correlation between all coloumns in data frame

correlation
```



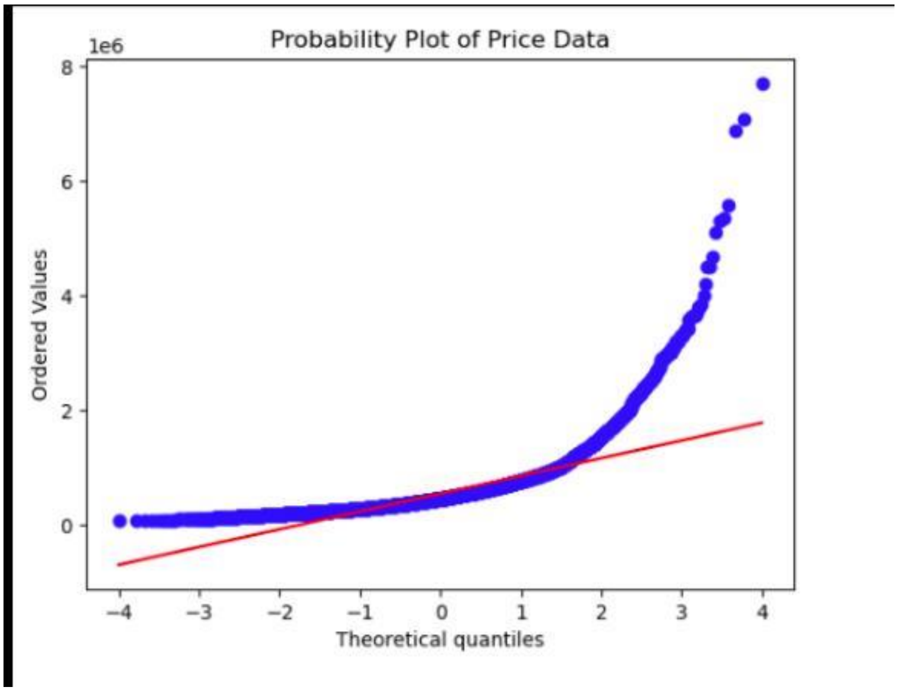
Overview of the Heatmap:

- The image consists of a grid of rectangles, each of which stands for two housing-related characteristics.
- The x- and y-axes show several property features, including price, lot size, living area square footage, number of bedrooms, and bathrooms.
- The color scale on the right side of the image displays the strength and direction of the relationship between these factors.
- Dark red hues suggest significant positive connections, while dark blue hues indicate strong negative correlations.

Correlation Strength:

- Each cell's color corresponds to the correlation coefficient between a given pair of variables.
- For example, a strong positive correlation between price and sqft_living (living space area) is indicated by the intense red hue.
- Similarly, further relationships can be inferred.

Probability Plot Overview



Overview of Probability Plot:

- On the graph, the pricing data is shown as a probability plot.
- A probability plot can be used to assess if a dataset (in this case, the normal distribution) closely resembles a specific theoretical distribution.
- It compares the observed data—actual values—with the theoretical quantiles, which stand in for the expected distribution, in order to identify deviations.

Graph components include:

- On the X-Axis, "Theoretical Quantiles":
  - ➔ The x-axis shows the representation of theory quantiles.

    - ➔ These quantiles originate from a theoretical distribution, like the normal distribution.
    - ➔ The x-axis typically shows values that range from negative to positive.

- Y-Axis ("Ordered Values"): The y-axis shows the ordered values in the dataset.
    - ➔ These values—which have been adjusted or scaled—are most likely the price.
    - ➔ The y-axis scale looks to be logarithmic, with 1e6 (1 million) as its greatest value.

Blue Dots:

- On the graph, the blue dots correspond to individual data points from the pricing dataset.
- Each dot stands for a different price.

Red Line:

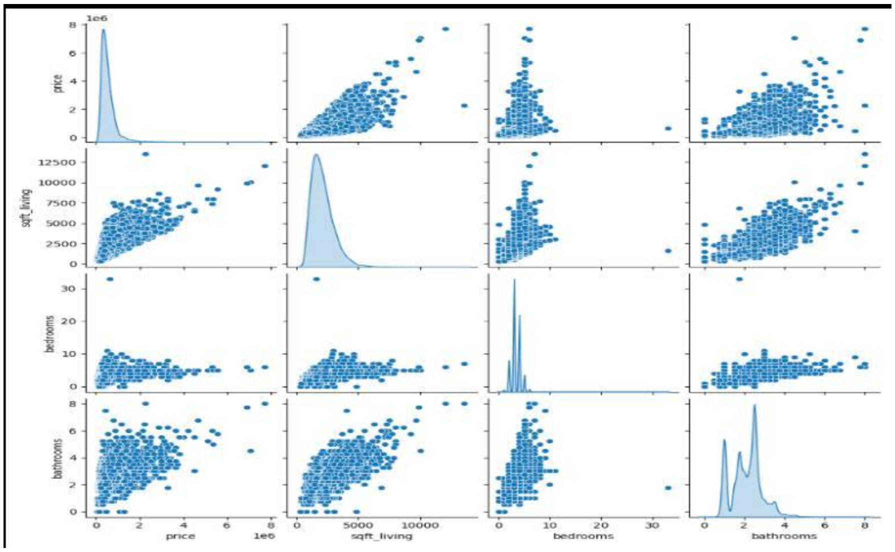
- The data would seem like the red line indicates if they were to follow a perfect normal distribution.
- It serves as a reference point.

Deviations of the blue dots from this line indicate non-normality.

Interpretation:

- If the blue dots follow the red line closely, then the data are generally regularly distributed.
- Deviations from the norm are implied by significant variances.
- The blue dots on the plot deviate from the red line.This disparity implies that the pricing data has a non-normal distribution.
- Non-normality can have an impact on statistical analyses that depend on normalcy, including regression models and hypothesis testing.

Pair plots between price,sqft_living,bedrooms and bathrooms :



One of the following variables is connected to each row and column:

- Cost: Displays the price of homes.
- Living Space Square Footage: This represents the living area's square footage.
- Number of Bedrooms: Indicates how many bedrooms each home has.
- Number of Bathrooms: Indicates how many bathrooms each home has.

The distribution of every variable is shown by these histograms:

- Cost: In a skewed distribution, most dwellings are found at the lower end.
- Living Space Square Foot: There is a similar imbalance in the distribution and pricing.
- A peak bedroom consists of three or four rooms.
- Bathrooms: A property's bathroom count is shown as a series of smaller peaks, centered around two bigger peaks.

Off-diagonal scatter plots: These graphs display bivariate correlations between two variable pairs.

Notable conclusions:

- Price and living space square footage have a positive link; as one rises, the other climbs as well.

- Though not as strongly as with each other, bedrooms and baths also exhibit positive correlations with cost and living area square footage.

Interpretation:

- The data points to a price-to-square-footage relationship between larger living areas and square footage.
- Though not as much, the number of bathrooms and bedrooms affects the cost as well.

MODEL ANALYSIS

Multiple Linear Regression

```
x1=data_redefined[[ 'bedrooms','bathrooms', 'sqft_living',
        'sqft_lot','floors','waterfront','view',
        'condition','grade', 'sqft_above','sqft_basement',
        'yr_built', 'yr_renovated','zipcode',
        'lat','long','sqft_living15','sqft_lot15']]
y1=data_redefined['price']
y1=sc.fit_transform(y1.values.reshape(-1, 1))

scr=[]
for i in range (100):
    train_x1,test_x1,train_y1,test_y1=train_test_split(x1,y1,test_size=0.1,random_state=i)
    # here test_size = 0.1 means 90% of data into train data and 10% of data into test data
    multi_regression=linear_model.LinearRegression()
    # Intialise the regression model using LinearRegression function imported from sklearn
    multi_regression.fit(train_x1,train_y1)
    # using fit function for fitting our model using train data
    scr.append(multi_regression.score(train_x1,train_y1))
    # Calculate score of the model using score function

scr # to view values in the list scr
```

```
print('The score of the model is : ',max(scr))
print('The random state is : ',scr.index(max(scr)))
print('The regression coefficients are : ',multi_regression.coef_ )
# finding coefficient of the best line equation
print('The intercept is : ',multi_regression.intercept_ )
# finding intercept of the best line equation
predicted_y1=multi_regression.predict(test_x1)
# Finding the predicted value of price
print('Mean squared error is : ',metrics.mean_squared_error(test_y1,predicted_y1))
# Finding the mean squared error incurred in the model
```

Output

```
The score of the model is :  0.7047205021746425
The random state is :  34
The regression coefficients are :  [[-9.75653888e-02  1.11882221e-01  3.02854766e-04  2.41758458e-07
   1.93198245e-02  1.65764467e+00  1.45582157e-01  7.04639443e-02
   2.62467947e-01  1.94765257e-04  1.08089509e-04 -7.16368574e-03
   6.31195626e-05 -1.62357580e-03  1.65278967e+00 -5.83944820e-01
   5.61353205e-05 -9.27142529e-07]]
The intercept is :  [20.03436902]
Mean squared error is :  0.26350881232591955
```

Output analysis

- Model Score: 0.7047 is the model's score.
  - ➔ This score indicates how well the regression model fit the data.
  - ➔ A higher score denotes a better fit.

- Random State: The selected random state is 34.
  - ➔ The random state is commonly employed as a seed for the generation of random numbers in order to ensure reproducibility.
  - ➔ Repeatedly running the model promotes consistency.

- Coefficients of Regression: The weights assigned to the different attributes (independent variables) within the dataset are displayed in these statistics.
  - ➔ These coefficients indicate the extent to which a one-unit change in each independent variable influences the dependent variable (price, for example), while all other variables are held constant.

- ➔ An array is used to represent the coefficients.

Interception:

- An intercept of 20.03436902 is present.
- The value of the predicted regression line where it crosses the y-axis is displayed when all x variables are zero.
- In essence, it is the basic value of the dependent variable.
- Mean squared error, or MSE has a value of 0.2635.
- The difference between the observed and estimated values is computed using the mean squared error, or MSE.
- Lower MSE values indicate better performance of the regression model.

Decision Tree Regressor

```
from sklearn.tree import DecisionTreeRegressor  # Import DecisionTreeRegressor
# Define the features and target variable
features = ['bedrooms', 'bathrooms', 'sqft_living', 'sqft_lot', 'floors', 'waterfront','view', 'condition', 'grade',
'sqft_above', 'sqft_basement', 'yr_built','yr_renovated', 'zipcode', 'lat', 'long', 'sqft_living15', 'sqft_lot15']
X = data[features]
y = data['price']

# Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.1, random_state=34)

# Scale the features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Simple Linear Regression
regression = linear_model.LinearRegression()
regression.fit(X_train_scaled, y_train)

# Model evaluation
train_score = regression.score(X_train_scaled, y_train)
print('The score of the model is:', train_score)

coefficients = regression.coef_
print('The regression coefficients are:', coefficients)
```

```
intercept = regression.intercept_
print('The intercept is:', intercept)

# Predictions
y_pred = regression.predict(X_test_scaled)

# Evaluate the model
mse = metrics.mean_squared_error(y_test, y_pred)
print('Men squared error is:', mse)
```

Output:

```
The score of the model is: 0.7047205021746425
The regression coefficients are: [-29704.5938592   28190.74019131  76793.96786648   6349.70719743
   7904.5139732   51131.9761045   39001.37525121  18044.40408262
 111950.17267859  70303.50028955  27449.36519656 -75637.89008164
   8588.94214689 -29024.99408053  83478.37329284 -28371.67027858
  19228.57829745 -10387.73116821]
The intercept is: 538650.0362963317
Mean squared error is: 63445448247.45971
```

Output analysis:

- Score of the Model: The model's score is around 0.7047.
  - ➔ This score indicates how well the regression model fit the data.
  - ➔ A higher score denotes a better fit.

- Random State: The selected random state is 34.
  - ➔ The random state is commonly employed as a seed for the generation of random numbers in order to ensure reproducibility.
  - ➔ Repeatedly running the model promotes consistency.

- Coefficients of Regression: The weights assigned to the different attributes (independent variables) within the dataset are displayed in these statistics.
  - ➔ These coefficients indicate the extent to which a one-unit change in each independent variable influences the dependent variable (price, for example), while all other variables are held constant.
  - ➔ An array is used to represent the coefficients.

Intercept:

- The approximate location of the intercept is 538650.
- The value of the predicted regression line where it crosses the y-axis is displayed when all x variables are zero.
- In essence, it is the basic value of the dependent variable.
- Mean squared error, or MSE is around 63445448247.
- MSE measures the difference between expected and actual performance.

RESULT

- We discovered that the Multiple Linear Regression model performs better than the Decision Tree Regressor when analyzing models for house price prediction.
- • Multiple Linear Regression Model: Score (R-squared): The model achieved an R-squared score of approximately 0.705.
- Mean Squared Error (MSE): The MSE is approximately 0.264, indicating relatively accurate predictions.
- • Decision Tree Regression: Score: The decision tree regression model achieved a similar R-squared score of approximately 0.705.
- Mean Squared Error (MSE): The MSE is significantly higher at approximately 63445448247.4597.
- Overall Comparison Both models perform similarly in terms of R-squared. However, the multiple linear regression model has a lower MSE, indicating better prediction accuracy

CONCLUSION

As indicated by its lower mean square error (MSE), the investigation's findings indicate that the Multiple Linear Regression model performed better than the Decision Tree model in predicting house values. This implies that the Multiple Linear Regression model generated more accurate forecasts than the Decision Tree model. This study highlights, therefore, how important it is to employ sophisticated modeling techniques in order to forecast real estate market developments.

The prediction power of these models can be enhanced in the future in a variety of ways. First off, adding larger and more diverse datasets could lead to more accurate and detailed predictions. By including additional variables, such as changes in the real estate market and socioeconomic indices, the models are able to account for a greater variety of factors influencing house prices. Policymakers and real estate investors may benefit from increased dependability in their decision-making processes if data sources are diversified.

Forecast accuracy may also be improved by looking into cutting-edge modeling techniques like gradient boosting machines and neural networks. These models may be better at predicting changes in the real estate market since they have shown to be more successful in a variety of domains. To ensure these approaches are robust and effective, it is necessary to assess their adaptability to different market situations.

In conclusion, additional study into complex modeling techniques and the inclusion of large datasets are required to enhance real estate prediction models. By applying cutting-edge techniques and expanding the breadth of their research, researchers can aid in the development of real estate market prediction models and assist in the making of smarter decisions. Ultimately, this will benefit all parties involved by providing more accurate and helpful insights into market dynamics and trends, especially legislators and real estate investors.

CITATIONS

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